Shuffle anti-squares of every even length from 24

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Abstract

A binary word is a shuffle square if it can be split into two identical subsequences, and an even word is a shuffle anti-square if none of its cyclic rotations is a shuffle square. Grytczuk, Pawlik and Pleszczyński found that the shortest shuffle anti-squares have length 24 and conjectured that they exist in every even length from 24 upward. We prove this. For every $k \geqslant 0$ the word $0^{k+9}\,1\,0^5\,11\,0^{k+9}\,1\,0^2\,1\,0\,111$ is a shuffle anti-square of length $2k + 34$, and the shorter lengths are covered by the family $0^{k+5}\,1\,0^2\,1111\,0^{k+4}\,111\,0\,1111$, which is in fact an anti-square for every k. The first family has a short human proof. Both families have been verified for all k in the Lean proof assistant. The proof for the first family extends to every word obtained from it by multiplying each zero run by a constant and replacing each one by the same odd number of ones. Since one cut can only rotate a word, some even binary words of every length from 24 need two cuts to become a shuffle square. The same authors and Ruciński found such a word of length 24 and conjectured that two cuts always suffice; by computer this holds up to length 34.

Mathematics Subject Classifications: 68R15, 05A05

1 Introduction

A word w is a *shuffle square* if its positions can be coloured with two colours so that the letters of each colour, read in order, spell the same word u. For example, 0110 is not a shuffle square: one copy must take the first 0 and one of the 1s, and spells 01, while the other spells 10. Its rotation 1100 is one, with copies 10 and 10. The question below is whether some rotation always rescues an even word in this way. Shuffle squares were introduced by Henshall, Rampersad and Shallit [7]. Deciding whether a binary word is a shuffle square is NP-complete [1], and He, Huang, Nam and Thaper conjectured that almost every binary word with an even number of each letter is one [5], and He and Post have since proved it [6].

Call a binary word *even* if it contains an even number of 0s and an even number of 1s. Grytczuk, Pawlik and Pleszczyński proved that every even binary word splits into two

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subwords, one of which is a cyclic shift of the other [2, Theorem 2], and conjectured that some cyclic rotation of every even binary word is a shuffle square, having checked this up to length 20 [3]. The conjecture fails at length 24. They called an even word a *shuffle anti-square* if none of its rotations is a shuffle square, found by computer that the shortest ones have length 24 (there is exactly one up to rotation, reversal and exchanging the letters), listed all anti-squares of lengths 24, 26 and 28, and made the following conjecture.

Conjecture 1 ([2, Conjecture 1])**.** There is a shuffle anti-square of length $2n$ for every $n \geqslant 12$.

We prove the conjecture with explicit families.

Theorem 2. *For $k \geqslant 0$ let*

$$U_k = 0^{k+9}\,1\,0^5\,11\,0^{k+9}\,1\,0^2\,1\,0\,111 \qquad \text{and} \qquad W_k = 0^{k+5}\,1\,0^2\,1111\,0^{k+4}\,111\,0\,1111.$$

Every U_k and every W_k is a shuffle anti-square. They have lengths $2k+34$ and $2k+24$, so Conjecture 1 holds.

Section 2 reduces the question to arithmetic on the runs of zeros. Section 3 proves that U_k is an anti-square by hand, in a stronger form (Theorem 4) that allows every zero run to be multiplied by a constant and every one to be replaced by an odd block of ones. For W_k the same reduction leaves a finite family of linear systems in the parameter k; Section 4 describes how these were checked, including a complete formalisation of Theorem 2 in Lean 4, and records counts of anti-squares up to length 34. Section 5 draws the consequence for the cutting distance of [3].

2 Runs of zeros

A binary word with m ones can be written uniquely as $w = 0^{e_0}10^{e_1}1\cdots10^{e_m}$; we call $(e_0,\dots,e_m)$ its *zero runs*. For example, 0110 has zero runs $(1,0,1)$. In a shuffle square the two colour classes are the *copies*.

Lemma 3. *Let $w = 0^{e_0}10^{e_1}1\cdots10^{e_m}$ be split into two copies A and B that both spell u. Then m is even, and u has zero runs $(r_0,\dots,r_{m/2})$ where, for each j, r_j equals the number of zeros of A lying between the j-th and $(j+1)$-th ones of A, and also equals the number of zeros of B lying between the j-th and $(j+1)$-th ones of B (with the conventions that the 0-th one is the start of w and the $(m/2+1)$-th one is its end).*

Proof. Each copy spells u, and the zero runs of a copy are read off exactly as stated. The ones of w split equally because both copies contain the ones of u. □

We shall use the lemma for rotations of a fixed circular word. Write the circular word as a cyclic sequence of ones separated by *gaps* of zeros. A rotation is determined by a cut point on the circle; the gap containing the cut contributes its two parts to e_0 and

e_m. Each copy's zeros in a gap all lie in the same zero run of that copy, namely the run spanning the gap.

It is convenient to merge the first and last zero runs of u into a single *cyclic run* $\rho_0 = r_0 + r_{m/2}$, and to write $\rho_j = r_j$ for $1 \leqslant j < m/2$. Reading around the circle from the cut, the cyclic runs of A and of B then agree term by term. This is weaker than Lemma 3, and we use the extra information only at the very end of Section 3: the cut lies inside the run ρ_0 of both copies, and the zeros of ρ_0 preceding the cut number $r_{m/2}$ in both copies.

3 The family U_k and its blow-ups

We prove a stronger statement, in which every zero run of U_k is multiplied by λ and every one is replaced by a block of μ ones. Read around a circle, let $V = V(K, \lambda, \mu)$ consist of

$$G = 0^{K\lambda}, \quad P_1 = 1^{\mu}, \quad g_1 = 0^{5\lambda}, \quad P_2 = 1^{2\mu}, \quad G' = 0^{K\lambda},$$
$$P_3 = 1^{\mu}, \quad g_2 = 0^{2\lambda}, \quad P_4 = 1^{\mu}, \quad g_3 = 0^{\lambda}, \quad P_5 = 1^{3\mu},$$

in this order. We call G and G' the *long gaps*, $C_1 = P_1 g_1 P_2$ and $C_2 = P_3 g_2 P_4 g_3 P_5$ the *clusters*, and the 8λ zeros of g_1, g_2, g_3 the *short* zeros. Read from the start of G, $V(k+9, 1, 1) = U_k$ (Figure 1).

Theorem 4. *If $K \geqslant 9$, $\lambda \geqslant 1$ and $\mu \geqslant 1$ is odd, then $V(K, \lambda, \mu)$ is a shuffle anti-square.*

Both conditions are needed: $V(8,1,1)$, $V(8,1,3)$, $V(9,1,2)$ and $V(9,1,4)$ are themselves shuffle squares (explicit splittings, checked in Lean). The proof uses $K > 8$ in the mass argument and in the final step, and the oddness of μ once, in the parity argument of Step 1. It never uses the parity of a number of zeros, which is why λ is arbitrary.

The proof has four steps. Comparing how the two copies share the long gaps, a mass count shows that the run of one copy through one long gap must coincide with the run of the other copy through the other long gap, or that one copy owns no one between them (Step 1). In the second case the fact that each copy holds half of the zeros already gives a contradiction (Step 2). In the first case, matching the remaining runs of the two copies index by index rules out every splitting but two (Step 3), and the position of the cut rules out those two (Step 4).

Suppose, for a contradiction, that some rotation of V is split into copies A and B spelling the same word u. By Lemma 3 each copy has 4μ ones, so u has 4μ cyclic runs, which we index modulo 4μ. Write $x_1, x_2, \alpha, \beta, \gamma$ for the numbers of zeros of G, G', g_1, g_2, g_3 in A, so that B has $y_1 = K\lambda - x_1$, $y_2 = K\lambda - x_2$, $5\lambda - \alpha$, $2\lambda - \beta$ and $\lambda - \gamma$ of them. Let a be the number of ones of A in C_1; then B has $3\mu - a$ ones in C_1 and $\mu + a$ in C_2, and exchanging the names of the copies replaces a by $3\mu - a$. For a copy X that has a one in C_2, let H_X and T_X be the numbers of zeros of X in C_2 before its first one and after its last one there; define H_X^1 and T_X^1 in the same way for C_1. The zeros of X in C_2 lying between two of its consecutive ones form its *inner runs* in C_2, and similarly for C_1. For example, if X owns exactly the ones of P_4 and P_5 in C_2, then H_X is its share of g_2,

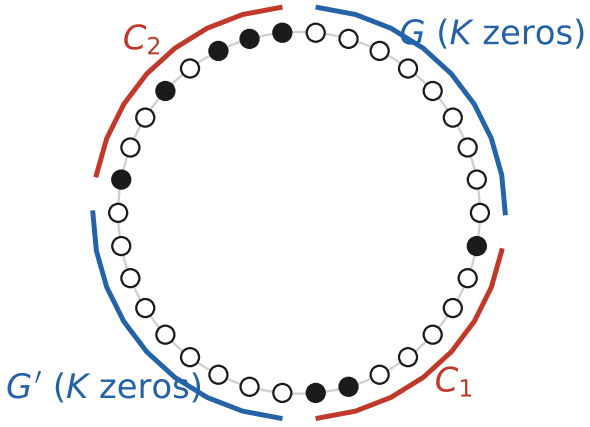


Figure 1: The anti-square $U_0 = V(9,1,1)$ read around a circle (filled: 1, open: 0). Two long gaps G and G' of $K = 9$ zeros separate the clusters $C_1 = 1\,00000\,11$ and $C_2 = 1\,00\,1\,0\,111$. The proof compares how two copies would share the zeros of the long gaps: the eight short zeros in the clusters allow only a small imbalance, inequality (2).

$T_X = 0$, and its only inner run in C_2 that can be nonempty is its share of g_3, between the last one of P_4 and the first one of P_5.

Two facts are used throughout. Both copies spell u, so each has half of the zeros:

$$x_1 + x_2 + \alpha + \beta + \gamma = (K+4)\lambda. \tag{1}$$

For each index j let $m_A(j)$ be the number of zeros from G and G' in the j-th cyclic run of A, define $m_B(j)$ likewise, and let $\sigma_A(j), \sigma_B(j)$ count the short zeros in these runs. Since $m_A(j) + \sigma_A(j) = \rho_j = m_B(j) + \sigma_B(j)$, summing over j gives

$$\sum_j |m_A(j) - m_B(j)| \leqslant 8\lambda. \tag{2}$$

3.1 Step 1: the long runs

If A's run spanning G has index i, its run spanning G' has index $i + a$, since A passes a of its ones in between; likewise B's two long runs have indices j and $j + 3\mu - a$.

Suppose $0 < a < 3\mu$, so that each copy has two distinct long runs. Two coincidences between an index of A and an index of B would force either $i = j$ and $2a \equiv 3\mu$, which is impossible because 3μ is odd and 4μ is even, or $j = i + a$ and $3\mu \equiv 0 \pmod{4\mu}$, also

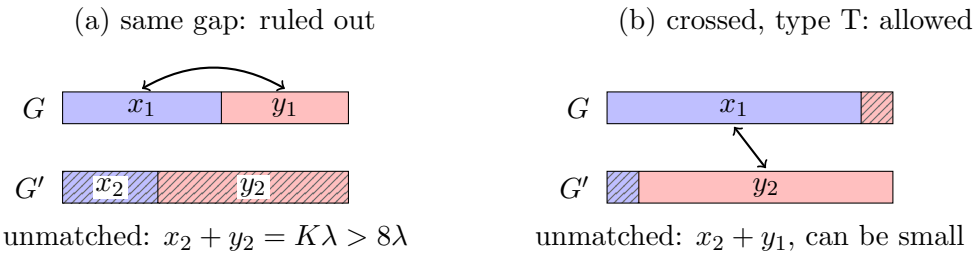


Figure 2: Step 1 in pictures. Each long gap is split between copy A (blue, shares x_1, x_2) and copy B (red, shares y_1, y_2); an arrow joins two shares that lie in runs with the same index, and hatched shares lie in runs that no long share of the other copy matches. By (2) the hatched total is at most 8λ. (a) If A's run through G lines up with B's run through G, then neither share of G' is matched, because A's long runs are a indices apart and B's are $3\mu - a$ apart, which never agree since 3μ is odd; that leaves $K\lambda$ zeros unmatched. (b) Lining up A's run through G with B's run through G' leaves only $x_2 + y_1$ unmatched, so this is the case the rest of the proof has to rule out.

impossible. With no coincidence the left side of (2) is $2K\lambda$, and when the coincidence is on the same long gap it is at least $K\lambda$; both exceed 8λ. So A's run spanning one long gap coincides with B's run spanning the other, and after exchanging the copies if necessary, A's run R spanning G coincides with B's run Q spanning G' (Figure 2). We call this *type T*.

If $a = 0$, the two long runs of A are a single run, and by (2) it coincides with one of the two long runs of B. If $a = 3\mu$, exchanging the copies returns us to $a = 0$.

3.2 Step 2: the case $a = 0$

Here B owns all of C_1 and $\mu \geqslant 1$ ones of C_2, and A's long run has $T_A + x_1 + \alpha + x_2 + H_A$ zeros. If it coincides with B's run spanning G', which has $y_2 + H_B = K\lambda - x_2 + H_B$ zeros, then (1) turns the equation into

$$T_A + H_A + x_2 + 4\lambda = \beta + \gamma + H_B.$$

The right side counts the zeros of C_2 held by A together with those of B before its first one there, so it is at most 3λ, a contradiction. If it coincides with B's run spanning G, with $T_B + y_1$ zeros, the same computation gives $T_A + H_A + x_1 + 4\lambda = \beta + \gamma + T_B \leqslant 3\lambda$, again a contradiction.

3.3 Step 3: type T

The run R has $T_A + x_1 + H_A^1$ zeros and Q has $T_B^1 + y_2 + H_B$. Equating them and using (1),

$$T_B^1 + H_B + \alpha + \beta + \gamma = 4\lambda + T_A + H_A^1. \tag{3}$$

Give R and Q the index 0. Going forward, A has its inner runs in C_1 at indices $1, \ldots, a-1$, its run X_2 spanning G', with $T_A^1 + x_2 + H_A$ zeros, at index a, and its inner runs in C_2

at $a+1,\dots,4\mu-1$. Likewise B has its inner runs in C_2 at $1,\dots,\mu+a-1$, its run Y_1 spanning G, with $T_B+y_1+H_B^1$ zeros, at index $\mu+a$, and its inner runs in C_1 at $\mu+a+1,\dots,4\mu-1$. Matching runs index by index:

(c1) at $1,\dots,a-1$, A's inner runs in C_1 equal B's first $a-1$ inner runs in C_2;

(c2) at a, X_2 equals B's inner run between its a-th and $(a+1)$-th ones in C_2;

(c3) at $a+1,\dots,\mu+a-1$, A's first $\mu-1$ inner runs in C_2 equal B's inner runs there;

(c4) at $\mu+a$, Y_1 equals A's inner run between its μ-th and $(\mu+1)$-th ones in C_2;

(c5) at $\mu+a+1,\dots,4\mu-1$, A's remaining inner runs in C_2 equal B's inner runs in C_1.

Sum (c1)–(c3) and (c3)–(c5), writing each total of inner runs as the zeros of that copy in the cluster less its zeros before the first and after the last one, and add the two equations. With F the total of the runs in (c3), this gives

$$H_A^1+T_B^1=2\lambda+H_B+2T_B+2H_A+T_A+x_2+y_1+2F. \tag{4}$$

Since C_1 begins with P_1 and ends with P_2, we have $H_A^1=0$ if A owns a one of P_1, and $H_A^1=\alpha$ otherwise; $T_B^1=0$ if B owns a one of P_2, and $T_B^1=5\lambda-\alpha$ otherwise. There are four branches.

If A owns a one of P_1 and B owns a one of P_2, the left side of (4) is 0, a contradiction. If A owns no one of P_1 but B owns a one of P_2, then (3) reads $H_B+\beta+\gamma=4\lambda+T_A$, while the left side is at most 3λ. If A owns a one of P_1 but B owns no one of P_2, then (3) reads $\lambda+H_B+\beta+\gamma=T_A$, while $T_A\leqslant\beta+\gamma$.

In the last branch A owns all of P_2 and none of P_1, so $a=2\mu$. Then A's inner runs in C_1 and B's inner runs in C_1 lie inside P_2 and P_1 and are empty, so (c1) says that B's first 2μ ones in C_2 have no zeros of B between them, and (c5) says the same of A's last μ ones in C_2. Let n_3,n_4,n_5 be the numbers of ones of A in P_3,P_4,P_5, so that $n_3+n_4+n_5=2\mu$, and B has $2\mu-n_3-n_4=n_5$ ones in $P_3\cup P_4$. In this branch $R=Q$ reads $T_A+x_1+\alpha=5\lambda-\alpha+y_2+H_B$, and eliminating α with (3) gives

$$H_B+2(\beta+\gamma)-T_A+x_2-y_1=3\lambda. \tag{5}$$

- If $1\leqslant n_5<\mu$, then B's first 2μ ones in C_2 include a one before g_3 and a one after it, so B has no zero of g_3 and $\gamma=\lambda$; but A's last μ ones also straddle g_3, so $\gamma=0$.

- If $n_5=2\mu$, then B owns P_3 and P_4, so (c1) gives $\beta=2\lambda$, and $H_A=\beta+\gamma$. By (c2), x_2+H_A equals B's run from the end of P_4 to its next one, which is $\lambda-\gamma$; that is, $x_2+2\lambda+2\gamma=\lambda$.

- If $\mu\leqslant n_5<2\mu$ and $n_3+n_4<\mu$, then (c1) again gives $\gamma=\lambda$. The runs in (c3) on B's side lie inside P_5, so A's first μ ones in C_2 have no zeros of A between them; they straddle g_3, so $\gamma=0$.

- If $n_3 + n_4 = \mu$ with $n_3, n_4 \geqslant 1$, then as before $\gamma = \lambda$ and A's first μ ones have no zeros of A between them, so $\beta = 0$; then (c4) gives $T_B + y_1 = \gamma$, so $y_1 = \lambda$, and (c2) gives $x_2 = H_A = 0$. Since B owns a one of P_3 and A a one of P_5, $H_B = T_A = 0$, and (5) reduces to $\lambda = 3\lambda$.

- If $n_3 = 0$ and $n_4 = \mu$, then B's first 2μ ones in C_2 run from P_3 into P_5, so $\beta = 2\lambda$; but (c2) forces $H_A = 0$, and $H_A = \beta$.

The remaining cases, $n_5 = 0$ and $(n_3, n_4) = (\mu, 0)$, give two splittings, and these satisfy every equation between cyclic runs:

(S1) A owns P_2, P_3 and μ ones of P_5; B owns P_1, P_4 and 2μ ones of P_5; here $\gamma = \lambda$, $x_2 = 0$ and, by (c4), $y_1 = \beta + \lambda$.

(S2) A owns P_2, P_3 and P_4; B owns P_1 and P_5; here $x_2 = 0$ and $y_1 = \beta$.

3.4 Step 4: the cut

By Lemma 3 the cut lies at a point of the circle covered by a run of A and the run of B with the same index, and the two copies have equal numbers of zeros of that run after the cut. In both (S1) and (S2) we have $y_1 \leqslant 3\lambda$, so $x_1 \geqslant (K-3)\lambda \geqslant 6\lambda$.

At index 0, R runs from A's last one in C_2 to the start of P_2, and Q from the end of P_1 to B's first one in C_2. They overlap in g_1, and in (S2) also in g_3. A cut in g_1 leaves before it all x_1 zeros of G in A's run but at most 5λ zeros in B's, so the counts after the cut differ as well. A cut in g_3 in (S2) leaves after it at most λ zeros of B, but at least x_1 zeros of A.

At the other indices, in (S2) the runs of A lie inside P_2 (indices 1 to $2\mu - 1$), span G' (index 2μ), lie inside P_3, span g_2 (index 3μ) and lie inside P_4, while those of B lie inside P_5 (indices 1 to $3\mu - 1$), span G (index 3μ) and lie inside P_1; no two with the same index meet. In (S1) the runs of A are the same up to index $3\mu - 1$, the run at 3μ goes from the end of P_3 to A's first one of P_5, and the later ones lie inside P_5; those of B lie inside P_4, then run from P_4 to its first one of P_5 (index μ), lie inside P_5, span G (index 3μ) and lie inside P_1. The only possible meeting is at index 3μ, when all of B's ones in P_5 precede all of A's; then the two runs share only the point between B's last one and A's first one of P_5. A cut at that point leaves no zero of A's run after it but $y_1 + T_B \geqslant y_1 = \beta + \lambda > 0$ zeros of B's.

This completes the proof of Theorem 4, and with $\lambda = \mu = 1$ and $K = k + 9$, the proof that U_k is a shuffle anti-square. □

4 The family W_k and computer verification

The words U_k cover the lengths $2n \geqslant 34$. For $24 \leqslant 2n \leqslant 32$ it suffices to exhibit one anti-square of each length, which is a finite check: Grytczuk, Pawlik and Pleszczyński list all of them for lengths 24, 26 and 28 [2], and W_3 and W_4 (lengths 30 and 32) pass

the same exhaustive test. The family W_k covers every length at once: W_0 is their unique anti-square of length 24, and W_k is an anti-square for every k.

For W_k the reduction of Section 2 leaves one system of linear equations and inequalities in k and the shares x_g for each cut gap and each way of dividing the twelve ones between the copies. There are 12×462 such systems, and none has a solution, even over the reals with $k \geqslant 0$. We checked this in three independent ways: by an exhaustive search over all rotations and splittings of W_k for $0 \leqslant k \leqslant 19$; with the SMT solver Z3 for the parametric systems; and in the Lean 4 proof assistant, where we defined interleavings and shuffle squares from scratch, proved Lemma 3 in the form of a splitting relation on zero runs, and discharged every case with the linear-arithmetic procedure `omega`. The Lean development proves, for every $n \geqslant 12$, that W_{n-12} is an even word of length $2n$ none of whose rotations is a shuffle square, and it depends on no axioms beyond propositional extensionality and quotient soundness. It also proves the statement for U_k independently of Section 3.

The exhaustive search reproduces the counts of [2] and extends them. Table 1 gives the number of anti-squares up to rotation, reversal and exchange of the letters.

length	24	26	28	30	32	34
anti-squares	1	26	103	660	2758	11026
fewest ones	12	10	10	10	8	8

Table 1: Shuffle anti-squares by length (equivalence classes), and the smallest possible number of the rarer letter.

Remark 5. Both families have two long gaps, each longer than all the short gaps together, separating two clusters with an odd number of ones. The parity argument of Section 3 uses only this, while the short gaps do the rest; the words $0^K1^50^K1^7$ and $0^K1^30^K1^5$, which have the same shape without short gaps, are not anti-squares for any $K \leqslant 20$ (by exhaustive check). It would be interesting to know the least number of ones in an anti-square of length $2n$ for large n, and whether it stays bounded.

5 Cutting distance

In another paper [3] the same authors measure how far an even word is from being a shuffle square by cuts. Cutting a word into pieces and rearranging them gives another word with the same letters; let $\mathrm{cut}(T, \mathcal{S}_2)$ be the least number of cuts that turns the even binary word T into a shuffle square, let $s(2, n)$ be its maximum over even binary words of length n, and let $s(2) = \sup_n s(2, n)$. They conjectured that $s(2) = 1$ [3, Conjecture 4], having checked $s(2, n) \leqslant 1$ for $n \leqslant 20$. With Ruciński they later showed that an anti-square of length 24 needs exactly two cuts, conjectured that two cuts always suffice, that is $s(2) \leqslant 2$, and noted that it is not even known whether $s(2)$ is finite [4, Section 6.3].

One cut turns $T = PQ$ into QP, a rotation, so $\mathrm{cut}(T, \mathcal{S}_2) \leqslant 1$ exactly when some rotation of T is a shuffle square. An anti-square therefore needs at least two cuts, and Theorem 2 extends their example of length 24 to every even length from 24.

Corollary 6. *$s(2, 2n) \geqslant 2$ for every $n \geqslant 12$.*

Their conjecture $s(2) \leqslant 2$ holds for every anti-square we know. Two cuts split T into XYZ, and apart from rotations the rearrangements are XZY, YXZ and ZYX. Testing these for every rotation of every anti-square of length at most 34, 1 938 732 words in all, found a shuffle square each time. For example, $W_0 = 0 \mid 0^4 1 \mid 0^2 1^4 0^4 1^3 0\, 1^4$ rearranges as $YXZ = 0^4 1\, 0^3 1^4 0^4 1^3 0\, 1^4$, whose two copies both spell $0^2 1 0^3 1^3 0\, 1^2$. For $4 \leqslant 2n \leqslant 22$ there are no anti-squares, so $s(2, 2n) \leqslant 1$, and the word $0110\, 0^{2n-4}$ is not itself a shuffle square: its first letter precedes both ones, so u begins with 0 and each copy needs a zero before its one, while only one zero precedes the ones. Hence $s(2, 2n) = 1$ for $4 \leqslant 2n \leqslant 22$ and $s(2, 2n) = 2$ for $24 \leqslant 2n \leqslant 34$. Whether $s(2)$ is finite remains open.

Code and formal proof

The programs, their outputs and the Lean 4 development are in the repository `https://github.com/jvvk/mathematics`, in the folder `shuffle-anti-squares`. The Lean development, built on definitions of interleavings, shuffle squares, rotations and cutting written from scratch, proves Theorem 2, Theorem 4 with K, λ and μ symbolic, Corollary 6, the identity $U_k = V(k + 9, 1, 1)$, the splittings showing that both conditions of Theorem 4 are needed, and the two-cut example of W_0 in Section 5. Its proof of Theorem 4 does not follow Section 3 line by line: for each of the ten runs in which the cut can fall, it splits into cases on which blocks of ones of the two copies overlap, and closes every case by linear arithmetic; a solver was used only to choose the order of the case splits. The development depends on no axioms beyond propositional extensionality, the axiom of choice and quotient soundness. Four statements rest on exhaustive computer search and are not formalised: the counts in Table 1, the absence of anti-squares below length 24, the claim that two cuts suffice up to length 34 in Section 5, and the check in Remark 5. Each was confirmed by a second program that shares no code with the first, and the counts up to length 28 also agree with the search of [2].

Acknowledgements

The families U_k and W_k were found by computer search. The author used the AI assistant Claude (Anthropic; Claude Opus 5.5 in October 2026 and earlier Claude models in September 2026) to write the search and verification programs, to explore the case analysis, to find and draft the proof of Theorem 4 and the observations of Section 5, and to produce the Lean formalisation, including the choice of case splits for Theorem 4. It was used to search quickly and to obtain machine-checked proofs: Theorems 2 and 4 and Corollary 6 are verified in Lean, as described above. The author takes full responsibility for the content of this paper.

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